

Probability HW6

Problem #1

Let weekly demand be $X_1, X_2 \sim \text{iid } N(\mu=100, \sigma^2=200^2)$

(a) $P(X_1 < 1100, X_2 < 1100)$

$$Z_i = \frac{X_i - \mu}{\sigma} = \frac{X_i - 1000}{200} \sim N(0, 1)$$

$$\begin{aligned} P(X_1 < 1100, X_2 < 1100) &= P(X_1 < 1100) \cdot P(X_2 < 1100) \\ &= P(Z_1 < 0.5) \cdot P(Z_2 < 0.5) \\ &= \Phi^2(0.5) \approx (0.69146)^2 \\ &\approx 0.4781 \end{aligned}$$

(b) $P(X_1 + X_2 > 2200)$

$$X_1, X_2 \sim \text{iid } N(\mu=1000, \sigma^2=200^2) \Rightarrow Y = X_1 + X_2 \sim N(2000, 80000)$$

$$Z_Y = \frac{Y - \mu_Y}{\sigma_Y} = \frac{Y - 2000}{200\sqrt{2}} \sim N(0, 1)$$

$$\begin{aligned} P(Y > 2200) &= P(Z_Y > \frac{1}{\sqrt{2}}) = 1 - P(Z_Y \leq \frac{1}{\sqrt{2}}) \\ &= 1 - \Phi(\frac{1}{\sqrt{2}}) \approx 1 - 0.76025 \\ &= 0.2398 \end{aligned}$$

Problem #2

$$\bar{X} = \frac{1}{100} \sum_{i=1}^{100} X_i$$

By CLT, for large n , $\bar{X} \sim N(\mu, \frac{\sigma^2}{n}) = N(2.2, \frac{0.3^2}{100}) = N(2.2, 0.0009)$

$$\mu(\bar{X}) = 2.2, \sigma(\bar{X}) = 0.03$$

$$P(\bar{X} > 3.1) = 1 - P(\bar{X} \leq 3.1) = 1 - P(\frac{\bar{X} - 2.2}{0.03} \leq \frac{3.1 - 2.2}{0.03}) = 1 - \Phi(30) \approx 4.9 \times 10^{-103} \approx 0$$

Problem #3

Let $X_i (i=1, 2, \dots)$ be iid lifetimes of the components

$$E(X_i) = \mu = 100, \text{SD}(X_i) = \sigma = 30, \text{total operating time} = Y_k = \sum_{i=1}^k X_i$$

by CLT, for large k , $Y_k = \sum_{i=1}^k X_i \sim N(k\mu, k\sigma^2) \Rightarrow \frac{Y_k - k\mu}{\sqrt{k}\sigma} \sim N(0, 1)$

$$P(Y_k \geq 2000) = P(\frac{Y_k - 100k}{30\sqrt{k}} \geq \frac{2000 - 100k}{30\sqrt{k}}) = 1 - \Phi(\frac{2000 - 100k}{30\sqrt{k}}) \geq 0.95$$

$$\Rightarrow P(\frac{2000 - 100k}{30\sqrt{k}}) \leq 0.05$$

$$\Rightarrow \frac{2000 - 100k}{30\sqrt{k}} \leq -1.645$$

$$\Rightarrow 100k - 2000 \geq 49.35\sqrt{k}$$

$$\text{let } t = \sqrt{k} \ (t \geq 0) \quad 100t^2 - 49.35t - 2000 \geq 0$$

let $100t^2 - 49.35t - 2000 = 0$
 $t_{1,2} = \frac{49.35 \pm \sqrt{49.35^2 + 4 \cdot 100 \cdot 2000}}{200}$ $t_1 = -4.232$, $t_2 = 4.726$
 $t \geq 0$ and $100t^2 - 49.35t - 2000 \geq 0$
 $t \geq 0$ and $100t^2 - 49.35t - 2000 \geq 0$
 $t \geq 4.726$ $k = t^2 \geq 22.34$
 k must be integer, $k = 23$

Problem #4

Let X_i ($i=1,2,\dots$) be iid dice rolls with $\mu = E(X_i) = \frac{1}{6}(1+2+3+4+5+6) = \frac{7}{2} = 3.5$
 $E(X_i^2) = \frac{1}{6}(1^2+2^2+3^2+4^2+5^2+6^2) = \frac{91}{6}$
 $\sigma^2 = \text{Var}(X_i) = E(X_i^2) - (E(X_i))^2 = \frac{35}{12}$
 let $Y = \sum_{i=1}^{400} X_i$ (discrete), require more than 140 rolls $\Rightarrow Y \leq 400$
 by CLT, $Y \sim N(\mu, 140\sigma^2) = N(490, \frac{1225}{3})$ $\mu_Y = 490$ $\sigma_Y = \sqrt{\frac{1225}{3}} \approx 20.2$
 $P(Y \leq 400) = P(\sqrt{\frac{3}{1225}}(Y-490) \leq \sqrt{\frac{3}{1225}}(400-490)) \approx \Phi(-4.4518) = 4.7 \times 10^{-6}$
 continuity correction:
 $P(Y \leq 400) = P(\sqrt{\frac{3}{1225}}(Y-490) \leq \sqrt{\frac{3}{1225}}(400.5-490)) \approx \Phi(-4.429) \approx 4.75 \times 10^{-6}$

Problem #5

Let $X \sim \text{Bin}(n=1000, p=0.25)$ be number of defective chips)
 $\mu = E(X) = np = 250$
 $\sigma = \text{Var}(X) = np(1-p) = 187.5$
 by CLT, $X \sim N(250, 187.5)$
 $P(X < 200) = P(X \leq 199) = P(\frac{Y-250}{\sqrt{187.5}} \leq \frac{199-250}{\sqrt{187.5}}) \approx \Phi(-3.724) \approx 9.9 \times 10^{-5}$
 continuity correction:
 $P(X < 200) = P(\frac{Y-250}{\sqrt{187.5}} \leq \frac{199.5-250}{\sqrt{187.5}}) \approx \Phi(-3.688) \approx 1.13 \times 10^{-4}$

Problem #6

$\hat{p}_m$ = sample proportion of men who rarely eat breakfast
 $\hat{p}_w$ = sample proportion of women who rarely eat breakfast
 By CLT, $\hat{p}_m \sim N(p_m, \frac{p_m(1-p_m)}{n_m})$, $\hat{p}_w \sim N(p_w, \frac{p_w(1-p_w)}{n_w})$
 $p_m = 0.455$ $\text{Var}(\hat{p}_m) = \frac{0.455 \times 0.545}{300} \approx 0.000827$ $SD(\hat{p}_m) \approx 0.0288$
 $p_w = 0.420$ $\text{Var}(\hat{p}_w) = \frac{0.420 \times 0.580}{300} \approx 0.000812$ $SD(\hat{p}_w) \approx 0.0285$
 $P(\hat{p}_w > \hat{p}_m) = P(\hat{p}_w - \hat{p}_m > 0)$
 let $D = \hat{p}_w - \hat{p}_m \sim N(p_w - p_m, \text{Var}(\hat{p}_w) + \text{Var}(\hat{p}_m))$

$$E(D) = p_w - p_m = 0.420 - 0.455 = -0.035$$

$$\text{Var}(D) = \text{Var}(\hat{p}_w) + \text{Var}(\hat{p}_m) = 0.000827 + 0.000812 = 0.001639$$

$$\text{SD}(D) \approx 0.0405$$

$$P(D > 0) = P\left(\frac{D + 0.035}{0.0405} > \frac{0 + 0.035}{0.0405}\right) = 1 - \Phi(0.864) \approx 1 - 0.806 = 0.194$$